



Weakly Supervised Semantic Segmentation for Lung Adenocarcinoma using Vision Transformers

Hammad Mir, Boguslaw Obara
School Of Computing, Newcastle University

Introduction

- Adenocarcinomas start in the cells that would normally secrete substances such as mucus. Belongs to non small cell lung cancer, accounting for more than 87 of approximate 47 000 annual cases in the UK^[1].
- Weakly Supervised Semantic Segmentation provides a well needed opportunity to determine pixel level labels given image level labels.
- Creating pixel level labels for histopathological images is expensive and time consuming and requires domain knowledge to properly determine the areas belonging to different classes.
- This Project focuses on exploring the use of Vision Transformer networks for Weakly supervised Semantic Segmentation of lung adenocarcinoma (WSSS4LUAD dataset^[2]).

Weakly Supervised Semantic Segmentation

- Weakly supervised semantic segmentation utilizes class activation maps to generate pseudo labels which are in-turn, after refinement, used to train a semantic segmentation model.
- Traditionally these methods use a trained convolutional neural networks and class activation maps to achieve this.
- CAM-based methods fall short as classification networks tend to identify objects from most discriminative features as opposed to complete object as required by semantic segmentation

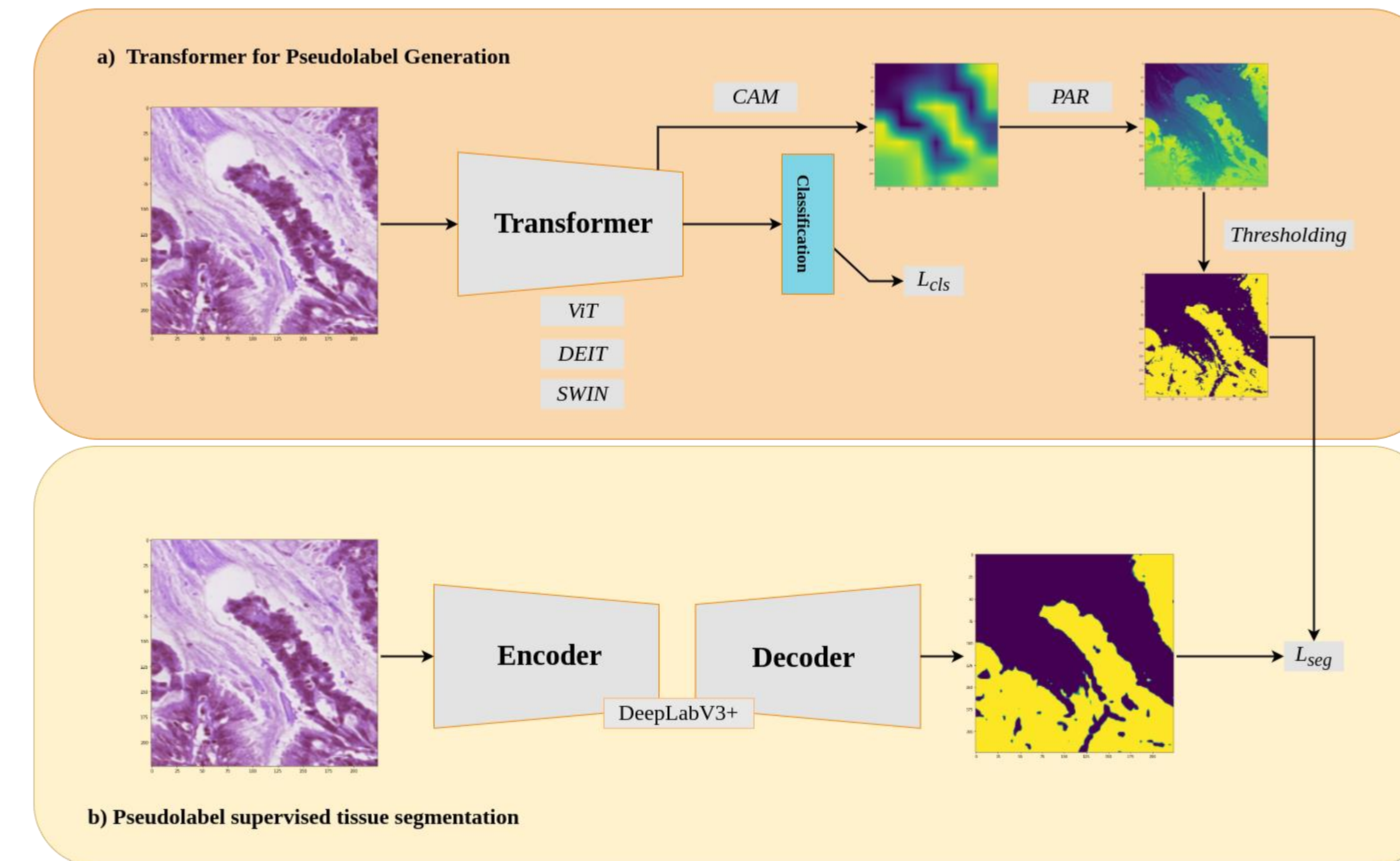
Related Work

- State-of-the-art performance has been achieved in tissue glands and nuclei localization^[3,4].
- Recently Vision Transformers have been gaining 'attention' for WSSS in natural images^[5,6].
- The DeepLab family of architectures pushed the limits of semantic segmentation, and with each network iteration, has produced cutting-edge results on benchmark datasets^[7].

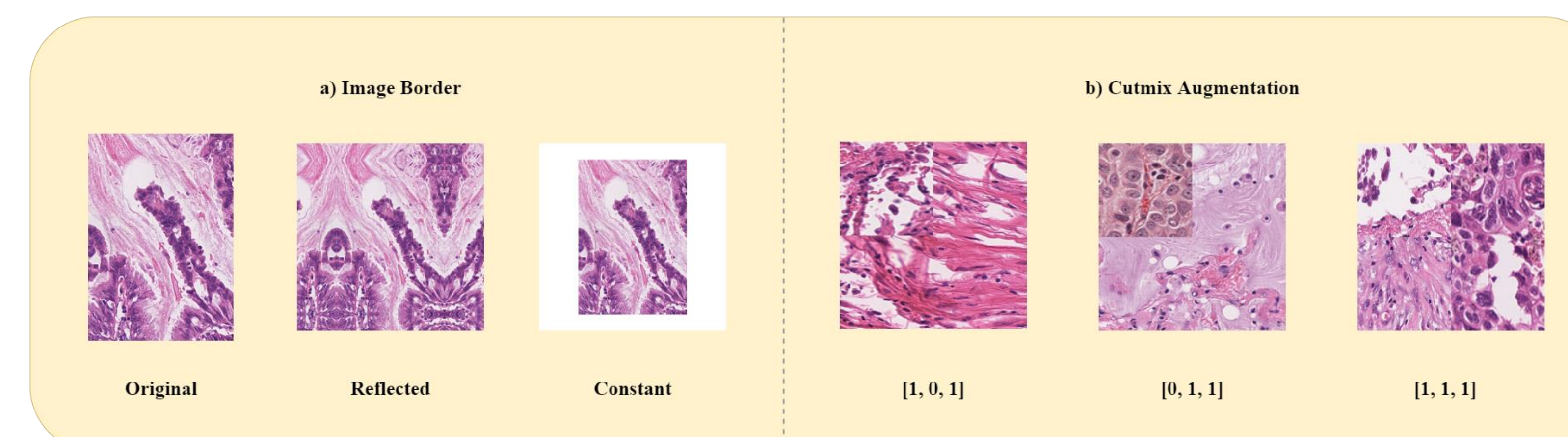
Methodology

Explain your results. Here are examples of figure formatting:

- We employed various Vision Transformer networks as classifier instead of traditionally used CNNs.
- We explored the use of various 'attention map' generation methodologies for vision transformers.
- The initial attention maps were refined using PAR and thresholding.
- These refined attention maps were used as training pseudo-labels to train the DeepLab V3++ segmentation model



Augmentation



- We augmented the images by adding a constant and reflected borders around the original images as the images vary in resolution.
- We also performed cutmix augmentation to address the missing multi-class data.

Results

The Table below shows the segmentation performance on the validation and the test datasets. The model struggles at distinguishing the tumour and tumour related stroma regions however the model performs significantly better at distinguishing the normal tissue from tumour as well as normal tissue for both validation and test sets.

Data	mIoU	Tumor	Stroma	Normal
Validation	0.58	0.58	0.38	0.78
	0.77	0.76		0.78
Test	0.63	0.53	0.42	0.94
	0.90	0.86		0.94

Discussion

- With this project we have established the feasibility of using vision transformer networks for WSSS in histopathology images.
- Our approach performs well at distinguishing the normal areas from tumor regions
- As evident from the results, the model struggles to clearly separate tumor from tumor associated stroma regions and further research is required to address this issue.

References

- [1] NHS UK Lung Cancer, <https://www.nhs.uk/conditions/lung-cancer/>, (08/2022)
- [2] Han, C., Pan, X., Yan, L., Lin, H., Li, B., Yao, S., Lv, S., Shi, Z., Mai, J., Lin, J. and Zhao, B., 2022. WSSS4LUAD: Grand Challenge on Weakly-supervised Tissue Semantic Segmentation for Lung Adenocarcinoma. arXiv preprint arXiv:2204.06455.
- [3] Van Rijnthoven, M., Balkenhol, M., Sillina, K., Van Der Laak, J. and Ciompi, F., 2021. HookNet: Multi-resolution convolutional neural networks for semantic segmentation in histopathology whole-slide images. Medical Image Analysis, 68, p.101890.
- [4] Han, C., Lin, J., Mai, J., Wang, Y., Zhang, Q., Zhao, B., Chen, X., Pan, X., Shi, Z., Xu, Z. and Yao, S., 2022. Multi-layer pseudo-supervision for histopathology tissue semantic segmentation using patch-level classification labels. Medical Image Analysis, p.102487.
- [5] Liu, C., Xie, E., Wang, W., Wang, W., Li, G. and Luo, P., 2022. WegFormer: Transformers for Weakly Supervised Semantic Segmentation. arXiv preprint arXiv:2203.08421.
- [6] Sun, W., Zhang, J., Liu, Z., Zhong, Y. and Barnes, N., 2021. GETAM: Gradient-weighted Element-wise Transformer Attention Map for Weakly-supervised Semantic segmentation. arXiv preprint arXiv:2112.02841.
- [7] Chen, L.C., Zhu, Y., Papandreou, G., Schroff, F. and Adam, H., 2018. Encoder-decoder with atrous separable convolution for semantic image segmentation. In Proceedings of the European conference on computer vision (ECCV) (pp. 801-818).